

Example 1. It has been established over many years of operation that the daily production from a certain small oil refinery is a normally-distributed variate with an average production of 19,425 barrels of oil per day (bbl/day) and a standard deviation of 500 bbl/day. (a) Determine the probability that on a random day the refinery will produce at most 19,680 bbl/day. (b) Determine the probability that on a random day the refinery will produce at least 19,680 bbl/day. (c) What is the probability that the oil refinery will produce between 19,000 and 20,000 barrels of oil on a randomly-selected day of operation? (d) What is the 95% confidence interval for daily oil production from the refinery?

Example 2. Assume that Li-ion cell phone batteries have an average lifetime (before failure) of 2.60 years with a standard deviation of 0.75 years. (a) Determine the probability that a randomly-selected battery will have a lifetime of at most 2.30 years. (b) Determine the probability that the average battery lifetime for a shipment of 25 batteries is at most 2.30 years.

Example 3. Early survey responses from 9 graduates of the 2011 BS ChE class at NCSU indicate the respondents have received an average salary of \$61,211 with a standard deviation of \$6,035. Based on the sample data: (a) What is the probability that the average salary of the entire 2011 BS ChE class (μ_x) will be below \$63,000? (b) What is the probability that μ_x will be above \$63,000? (c) What is the probability that μ_x will fall in the \$60,000 - \$63,000 range? (d) What is the 95% confidence interval for μ_x ?
